不可压缩 Navier–Stokes 方程及相关模型研究的数学工具

代表性试译样本

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1 流体力学方程

为简单起见, 在叙述和说明本章结果时, 我们对用于刻画流动的变量 (密度, 速度等) 的正则性作出若干假设. 不过应注意, 在某些情形下可以放宽这些假设.

1.1 输运定理

我们首先叙述一个基本的微分结果, 在流体力学中称为输运定理. 它说明如何对流体元 $(\Omega_t)_t$ 上积分的标量值求导.

我们假设, 对每个 t, 映射 φ_t 是光滑微分同胚, 并且 $t \mapsto \varphi_t$ 是光滑的. 为略微简化记号, 还令

$$\varphi(t,x) = \varphi_t(x) = X(t,0,x).$$

定理 1.1:
设函数 f 关于变量 $(t,X) \in \mathbb{R} \times \mathbb{R}^3$ 属于 C^1 类. 则对任意时刻 t, 有

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{\Omega_t} f(t,X)\,\mathrm{d}X = \int_{\Omega_t}\left(\frac{\partial f}{\partial t} + \operatorname{div}(fv)\right)\mathrm{d}X,$$

其中 v 是由前述运动学关系定义的 $\mathbb{R}^3$ 中的向量场.

证明: 固定 t. 作变量替换 $x \in \Omega_0 \mapsto X = \varphi(t,x) \in \Omega_t$, 得

$$\int_{\Omega_t} f(t,X)\,\mathrm{d}X = \int_{\Omega_0} f(t,\varphi(t,x))|J(t,x)|\,\mathrm{d}x,$$

其中 J 是映射 $x \mapsto \varphi(t,x)$ 的 Jacobian 行列式.

此外, 行列式关于其三行是三线性函数. 由前述等价关系, 推得恒等式

$$\frac{\partial}{\partial t}\frac{\partial \varphi_i}{\partial x_j}(t,x) = \frac{\partial}{\partial x_j}\frac{\partial \varphi_i}{\partial t}(t,x) = \frac{\partial}{\partial x_j}\big(v_i(t,\varphi(t,x))\big).$$

省略导数取值点的书写, 得

$$\frac{\mathrm{d}}{\mathrm{d}t}J(t,x) = \begin{vmatrix} \frac{\partial v_1}{\partial x_1} & \frac{\partial v_1}{\partial x_2} & \frac{\partial v_1}{\partial x_3} \\ \frac{\partial \varphi_2}{\partial x_1} & \frac{\partial \varphi_2}{\partial x_2} & \frac{\partial \varphi_2}{\partial x_3} \\ \frac{\partial \varphi_3}{\partial x_1} & \frac{\partial \varphi_3}{\partial x_2} & \frac{\partial \varphi_3}{\partial x_3} \end{vmatrix} + \begin{vmatrix} \frac{\partial \varphi_1}{\partial x_1} & \frac{\partial \varphi_1}{\partial x_2} & \frac{\partial \varphi_1}{\partial x_3} \\ \frac{\partial v_2}{\partial x_1} & \frac{\partial v_2}{\partial x_2} & \frac{\partial v_2}{\partial x_3} \\ \frac{\partial \varphi_3}{\partial x_1} & \frac{\partial \varphi_3}{\partial x_2} & \frac{\partial \varphi_3}{\partial x_3} \end{vmatrix} + \begin{vmatrix} \frac{\partial \varphi_1}{\partial x_1} & \frac{\partial \varphi_1}{\partial x_2} & \frac{\partial \varphi_1}{\partial x_3} \\ \frac{\partial \varphi_2}{\partial x_1} & \frac{\partial \varphi_2}{\partial x_2} & \frac{\partial \varphi_2}{\partial x_3} \\ \frac{\partial v_3}{\partial x_1} & \frac{\partial v_3}{\partial x_2} & \frac{\partial v_3}{\partial x_3} \end{vmatrix}.$$

应用链式法则, 得

$$\frac{\partial}{\partial x_j}\big(v_k(t,\varphi(t,x))\big) = \sum_{\ell=1}^{3}\frac{\partial v_k}{\partial X_\ell}(t,\varphi(t,x))\frac{\partial \varphi_\ell}{\partial x_j}(t,x). \tag{1.1}$$

定义行向量 L_k 和 M_k 如下:

$$L_k = \left(\frac{\partial \varphi_k}{\partial x_1}(t,x), \frac{\partial \varphi_k}{\partial x_2}(t,x), \frac{\partial \varphi_k}{\partial x_3}(t,x)\right),$$

$$M_k = \left(\frac{\partial}{\partial x_1}\big(v_k(t,\varphi(t,x))\big), \frac{\partial}{\partial x_2}\big(v_k(t,\varphi(t,x))\big), \frac{\partial}{\partial x_3}\big(v_k(t,\varphi(t,x))\big)\right).$$

式(1.1)表明

$$M_k = \sum_{\ell=1}^{3} \frac{\partial v_k}{\partial X_\ell}(t, \varphi(t,x)) L_\ell,$$

因而

$$\begin{aligned}
\det(M_1, L_2, L_3) &= \det\left(\sum_{\ell=1}^{3} \frac{\partial v_1}{\partial X_\ell} L_\ell, L_2, L_3 \right) \\
&= \sum_{\ell=1}^{3} \frac{\partial v_1}{\partial X_\ell} \det(L_\ell, L_2, L_3) \\
&= \frac{\partial v_1}{\partial X_1} \det(L_1, L_2, L_3) = \frac{\partial v_1}{\partial X_1}(t, \varphi(t,x)) J,
\end{aligned}$$

这是因为行列式是三线性且交错的. 类似计算给出

$$\det(L_1, M_2, L_3) = \frac{\partial v_2}{\partial X_2}(t, \varphi(t,x)) J, \qquad \det(L_1, L_2, M_3) = \frac{\partial v_3}{\partial X_3}(t, \varphi(t,x)) J.$$

另一方面, 已知

$$\frac{\partial J}{\partial t} = \det(M_1, L_2, L_3) + \det(L_1, M_2, L_3) + \det(L_1, L_2, M_3).$$

所以

$$\frac{\partial J(t,x)}{\partial t} = (\operatorname{div} v)(t, \varphi(t,x)) J(t,x).$$

我们已经假定 $\varphi(t)$ 是连续依赖于 t 的 C^1 微分同胚, 因而其 Jacobian 行列式 J 不为零, 且其值全都位于 $\mathbb{R}_*^+$ 或全都位于 $\mathbb{R}_*^-$ 中. 在这些区域上, "绝对值" 函数为 C^∞, 且导数为 "符号" 函数. 因此, 函数 $|J|$ 可微, 并且

$$\frac{\partial |J|}{\partial t} = \operatorname{sgn}(J) \frac{\partial J}{\partial t} = \operatorname{sgn}(J)(\operatorname{div} v)(t, \varphi(t,x)) J = (\operatorname{div} v)(t, \varphi(t,x)) |J|.$$

应用积分号下求导理论, 得

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t} \int_{\Omega_t} f(t,X) \, \mathrm{d}X &= \frac{\mathrm{d}}{\mathrm{d}t} \int_{\Omega_0} f(t, \varphi(t,x)) |J(t,x)| \, \mathrm{d}x \\
&= \int_{\Omega_0} \frac{\partial}{\partial t} \big(f(t, \varphi(t,x)) |J(t,x)| \big) \, \mathrm{d}x.
\end{aligned}$$

此外,

$$\begin{aligned}
&\frac{\partial}{\partial t} \big(f(t, \varphi(t,x)) |J(t,x)| \big) \\
&\quad = \frac{\partial f}{\partial t}(t, \varphi) |J| + \left(\frac{\partial \varphi}{\partial t} \right) \cdot \nabla f \, |J| + f(t, \varphi) \frac{\partial}{\partial t} |J| \\
&\quad = \left(\frac{\partial f}{\partial t}(t, \varphi) + v \cdot \nabla f + f(t, \varphi)(\operatorname{div} v)(t, \varphi(t,x)) \right) |J|.
\end{aligned}$$

对上式作反向变量替换, 得

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t} \int_{\Omega_t} f(t,X) \, \mathrm{d}X &= \int_{\Omega_t} \left(\frac{\partial f}{\partial t} + v \cdot \nabla f + f \operatorname{div} v \right) \mathrm{d}X \\
&= \int_{\Omega_t} \left(\frac{\partial f}{\partial t} + \operatorname{div}(fv) \right) \mathrm{d}X.
\end{aligned}$$

■

注 1.2: 这个标量方程的微分结果可以推广到向量方程. 为方便起见, 我们在下方以简洁形式给出此结果, 其中 $F(t,X)$ 是向量:

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{\Omega_t} F(t,X)\,\mathrm{d}X = \int_{\Omega_t}\left(\frac{\partial F}{\partial t} + \mathrm{div}(F\otimes v)\right)\,\mathrm{d}X,$$

这里的向量场 v 与定理1.1中的相同. 关于 $\mathrm{div}(F\otimes v)$ 的定义, 参见附录中的向量算子说明.